Achieve!

Chemistry in cooking

Baking soda is a chemical compound used when cooking cakes. The chemical name for baking soda is **sodium hydrogen carbonate**. It has the chemical formula $NaHCO_3$. Baking soda is used in baking because when it is heated, it breaks down to form sodium carbonate, water and carbon dioxide gas. The carbon dioxide gas expands, making the cake rise. The trapped carbon dioxide gas makes the cake light and fluffy.

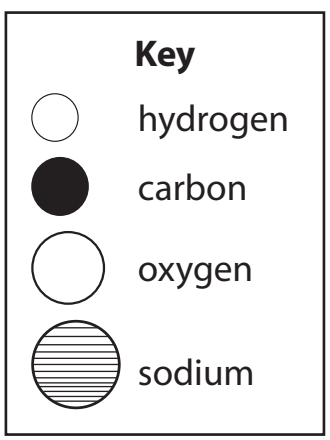
☞ **1** Complete the table below.

Common name	Chemical name	Chemical formula	Elements present
Baking soda			

2 Match each element to the number of atoms present in a molecule of baking soda.

sodium	carbon	hydrogen	oxygen
1	1	3	1

3 Using the key provided, draw a diagram of a molecule of baking soda.



4 Heating baking soda is a **decomposition** reaction. This means that one chemical breaks down into two or more simpler substances. How many substances does baking soda break down into when heated? _____

5 Name these substances. _____

6 Complete the word equation for the decomposition of baking soda (sodium hydrogen carbonate).

_____ $\xrightarrow{\text{heat}}$ _____ _____ + _____ + _____ _____

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